1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Coding:

```
import pandas as pd


data = pd.DataFrame([
   ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
   ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
   ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
   ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
], columns=['Sky', 'Temp', 'Humidity', 'Wind',
         'Water', 'Forecast', 'EnjoySport'])

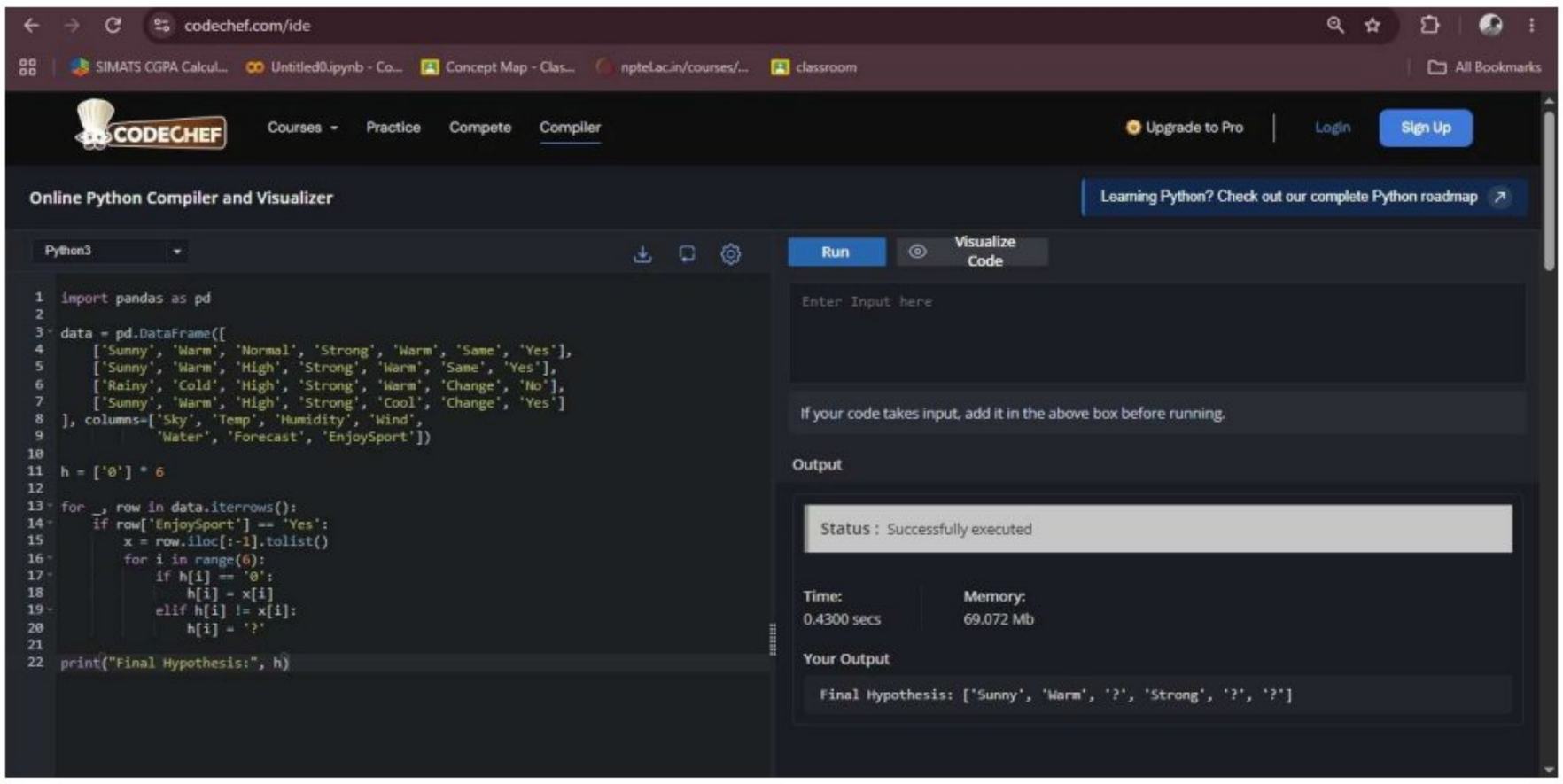

h = ['0'] * 6


for _, row in data.iterrows():
   if row['EnjoySport'] == 'Yes':
      x = row.iloc[:-1].tolist()
      for i in range(6):
         if h[i] == '0':
            h[i] = x[i]
         elif h[i] != x[i]:
            h[i] = '?'


print("Final Hypothesis:", h)
```

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OUTPUT:



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